Review I Report

Project title: Diabetes Prediction System
Technical focus: Binary diabetes-outcome classification with an artificial neural network (ANN)
Review: Review I
Review date: 8 October 2026
Classroom dataset: Pima Indians Diabetes (local file already in this workspace)

> **Disclaimer.** This is an educational machine-learning project, not a medical diagnostic tool, screening app, or clinical decision system. The dataset is small (768 rows) and limited to adult women (age 21+) of Pima heritage near Phoenix, Arizona. Results do not generalize to other populations, to men, or to children, and must not be used to diagnose diabetes.

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Class: AIML — "A" — 3rd year

This package covers the **six visible Review I criteria (40 marks)**. No additional rubric rows were provided in the request or found in the project files.

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1. Problem Definition and Objectives (5 marks)

1.1 Problem

Diabetes mellitus is a chronic condition in which blood glucose is poorly controlled. In clinical practice, diagnosis uses laboratory criteria and a clinician's judgement. In this classroom project the task is narrower and fully specified:

Given eight numeric measurements from one historical table, classify each row as Outcome = 1 (tested positive for diabetes by the study's rule) or Outcome = 0 (tested negative).

The learning problem is therefore **binary classification**. The model is an **artificial neural network**: a small feed-forward multilayer perceptron. It is compared with logistic regression and a random forest so that any ANN result can be judged against simple, well-understood baselines.

The table used here is the **Pima Indians Diabetes** extract associated with Smith et al. (1988) and later distributed as a machine-learning benchmark. In the original study the eight inputs were used to **forecast five-year onset** of diabetes in Pima women who were not diabetic at the index examination. The public 768-row file is a classroom benchmark derived from that work. It is not a contemporary, mixed-sex, multi-country clinical registry.

1.2 Motivation

Two motivations are kept separate on purpose.

Public-health context (why diabetes matters). The World Health Organization reports that the number of people living with diabetes rose from 200 million in 1990 to 830 million in 2022, and that adult prevalence roughly doubled from 7% to 14% over that period. Those figures describe a global health problem. They do **not** justify treating this student model as a diagnostic device.

Technical context (why this project exists). Undergraduate ANN courses need a small, fully observed tabular problem in which students can:

- inspect messy real measurements (coded zeros, imbalance);
- fit a neural net without a GPU;
- compare it fairly with classical models;
- practise evaluation beyond accuracy.

The Pima table is widely used for that purpose. It is also easy to misuse: many student reports scale or select features on all 768 rows, treat every zero as missing (or none), and quote accuracy near 65–80% as if that were clinical performance. This project is motivated by doing the classroom task **honestly**.

1.3 Aim

To design, document, and evaluate a **leakage-safe** pipeline that trains a small feed-forward ANN to classify the binary diabetes outcome on the local Pima Indians Diabetes table, and to compare that ANN with logistic regression and a random-forest baseline using stratified cross-validation and a full set of classification metrics.

1.4 Measurable objectives

ID	Objective	How completion is checked
O1	Load the dataset from a **local** CSV (or convert the local ARFF). Do not download data inside the code.	File path documented; missing-file error prints obtain/placement instructions.
O2	Profile class balance, duplicates, true missing values, and zero counts.	Written data profile with counts.
O3	Recode implausible zeros as missing only where justified; keep `Pregnancies = 0`.	Zero-handling table in this report and in code.
O4	Fit imputation, scaling, feature selection, and any resampling **only on each training fold**.	scikit-learn / imbalanced-learn pipeline; unit tests.
O5	Produce EDA plots: outcome balance, feature distributions by outcome, boxplots, correlation heatmap.	Figures under `diabetes_ann/reports/figures/`.
O6	Compare all eight features versus `SelectKBest` ($k=5$) inside each training fold.	Per-fold selected names recorded.
O7	Train a small MLP (hidden layers 16 and 8) and the two baselines under the same folds.	Same CV protocol for every model.
O8	Report accuracy, precision, recall/sensitivity, specificity, F1, ROC-AUC, PR-AUC, and confusion matrices.	Metric table; no claim of “high performance” as a clinical result.

1.5 Scope

In scope

- Binary classification on the 768-row Pima classroom table.
- A small feed-forward ANN (multilayer perceptron).
- Fair baselines: dummy most-frequent (sanity floor), logistic regression, random forest.
- Leakage-safe preprocessing and fold-wise feature selection.
- Educational interpretation of metrics and limitations.

Out of scope

- Medical diagnosis, screening, or treatment advice.

- External validation on other hospitals, sexes, ages, or ethnic groups.
- Deep architectures, transfer learning, or large language models.
- Hyperparameter search on the test fold.
- Claiming that the ANN is “better than doctors.”

1.6 Research questions

1. After leakage-safe preprocessing, how does a small feed-forward ANN compare with logistic regression and a random forest on this table when sensitivity, specificity, F1, ROC-AUC, and PR-AUC are all reported?
2. Which of the eight predictors does `SelectKBest` keep inside each training fold, and does using $k=5$ change those metrics relative to using all features?
3. How should zeros be handled so that medically implausible laboratory zeros are not treated as true measurements, while `Pregnancies = 0` remains valid?

1.7 Status of work (not a performance claim)

The local CSV is present. Data profiling, EDA plots, the preprocessing pipeline, and a seeded 5-fold comparison have already been run in this workspace. **Review I uses those facts for dataset understanding, preprocessing, EDA, and feature-selection design.** Classifier scoreboards belong to later reviews if the unseen rubric asks for them. Numbers quoted below for class counts, zeros, correlations, and selected features are **computed from the local file**, not invented.

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2. Literature Survey and Research Gap (8 marks)

The survey is limited to sources that could be checked. Paper details that were not available in an open full text are marked. Citations and link-check notes are in `references.md`.

2.1 Original study and what the public table actually is

Smith, Everhart, Dickson, Knowler, and Johannes (1988) introduced the modelling task. They used an early connectionist algorithm called **ADAP** to forecast five-year onset of non-insulin-dependent diabetes in Pima women near Phoenix. NIDDK had followed that community since 1965. Diabetes was defined by WHO-style rules: 2-hour post-load plasma glucose \$geq 200$ mg/dL (11.1 mmol/L) at a survey examination, or a glucose of at least 200 mg/dL found in routine care.

Eight inputs were used (pregnancies, 2-hour OGTT glucose, diastolic blood pressure, triceps skinfold, 2-hour serum insulin, BMI, a diabetes pedigree function, and age). The pedigree function was a constructed summary of relatives’ diabetes history; it is **not** a laboratory assay.

Evaluation (verified from the paper/PMC text): 768 examinations met the inclusion rules; 576 were used for training and 192 for forecasting. Forcing a decision on every test case gave **sensitivity and specificity of 76%**. An ROC curve was reported; the sensitivity–specificity crossover was 0.76. The authors stated they were comparing ADAP with logistic regression and a linear perceptron on the **same** splits.

Limits (from the same paper): female subjects only; age at least 21; one examination per person; examinations that became diabetic within one year were excluded because they were considered too easy. The public CSV inherits those constraints. It is a forecast-style extract from a high-risk community, not a random sample of “patients everywhere.”

Knowler et al. (1978), cited by Smith et al., documented a much higher diabetes incidence in Pima adults than in Rochester, Minnesota. That explains why NIDDK studied this community. It is **not** a reason to deploy a 768-row

student ANN as a general diagnostic model.

2.2 Neural-network work on the same UCI extract

Temurtas, Yumusak, and Temurtas (2009) compared a multilayer neural network trained with Levenberg–Marquardt and a probabilistic neural network on the UCI Pima extract. They used a conventional train/test split and 10-fold cross-validation, and compared accuracy with earlier Pima neural-network papers. **Verified from the abstract and publisher pages.** Exact per-model percentages are not copied here from secondary summaries.

Limit often discussed for this literature: several neural-network Pima papers emphasise **accuracy**. Bozkurt et al. (2014), comparing earlier Pima classifiers including Temurtas et al., note that most of those studies reported accuracy only and omitted sensitivity and specificity. That matters because 500 of 768 rows are negative: a majority-class rule already scores about 65% accuracy.

2.3 Classical classifiers on Pima

Sisodia and Sisodia (2018) trained decision tree, SVM, and naïve Bayes on the Pima Indians Diabetes Database from UCI. They reported precision, accuracy, F-measure, recall, and ROC curves, and stated that naïve Bayes reached **76.30% accuracy**, the best of the three. **Verified from the publisher/ACM abstract.** The abstract does not fully specify whether preprocessing and any attribute selection were nested inside cross-validation. That protocol detail is therefore **not claimed here**.

2.4 Broader ML pipelines, including MLP

Hasan, Alam, Das, Hossain, and Hasan (2020) (IEEE Access) used the Pima Indian Diabetes Dataset with outlier handling, filling missing values, standardisation, feature selection, *k*-fold cross-validation, several classifiers **including MLP**, and an AUC-weighted ensemble. They report, for the proposed ensemble, sensitivity 0.789, specificity 0.934, and AUC 0.950. **Verified from the article abstract / author summary.** IEEE full-text fetch was blocked in this environment; the DOI is recorded in `references.md`.

An AUC of 0.95 on 768 mixed-quality rows is a strong published number. For an undergraduate review it is a **caution**, not a target to beat by leaking test information. Outlier rejection and feature selection can inflate scores if they are fitted on the full table.

Howlader et al. (2022) (open access) used the Kaggle copy of Pima, several feature-selection methods, and many classifiers. They report high accuracy for some models (abstract: Generalized Boosted Regression 90.91%) and state that glucose, BMI, diabetes pedigree function, and age were consistently important. **Verified from the PMC full text.** Such accuracy on this extract is unusually high relative to the original 76% sensitivity/specificity operating point and to many later Pima papers in the mid-70% accuracy range. This project does not try to reproduce 90%+ figures.

2.5 Data-quality literature (zeros are not “no missing values”)

The historical UCI-style documentation says **“Missing Attribute Values: None.”** That statement is about the file format: empty cells were not used. The mlbench R documentation (Leisch and Dimitriadou) states the important correction: zeros in glucose, blood pressure, triceps, insulin, and BMI are physical impossibilities, and in `PimaIndiansDiabetes2` those zeros are set to `NA`, following Wahba et al. (1995) and Ripley (1996).

This project follows that correction **except** that pregnancies are left as zero, because zero pregnancies is valid.

2.6 Comparison table

Source	Dataset	Methods	Evaluation (as verified)	Limitation
Smith et al., 1988	Pima women, 8 inputs, 768 exams, 5-year onset	ADAP (early neural / perceptron-like)	576 / 192 split; sensitivity = specificity = 76%; ROC	One high-risk female community; not a general diagnostic model
Temurtas et al., 2009	UCI Pima extract	MLNN (LM) and PNN	Conventional split and 10-fold CV; accuracy vs prior NN papers	Abstract centres on accuracy; full metric set not verified here from full text
Sisodia & Sisodia, 2018	UCI PIDD	Decision tree, SVM, naïve Bayes	Precision, recall, F-measure, accuracy; NB 76.30%; ROC	Nested vs non-nested preprocessing not verified from abstract
Hasan et al., 2020	Pima	Cleaning, selection, *k*-fold, MLP and ensembles	Sensitivity, specificity, AUC (ensemble AUC 0.950 in abstract)	Very high AUC on a small dirty table; leakage risk if selection is global
Howlader et al., 2022	Kaggle Pima	Many selectors and classifiers	Accuracy, AUROC, sensitivity, specificity, and others	Some reported accuracies are far above typical Pima baselines
mlbench / Ripley / Wahba	Same extract	Documentation, not a new classifier	Shows UCI "no missing values" is misleading	Not a modelling paper

2.7 Research gap (modest and specific)

This is an undergraduate project. The gap is **not** "nobody has trained a neural net on Pima." Smith et al. already did, and many papers followed.

The **modest gap** this project addresses is:

> On this classroom benchmark, a large share of student (and some published) pipelines still (i) treat the UCI "no missing values" note as literal, or recode **all** zeros including pregnancies; (ii) fit scalers, imputers, or feature selectors on **all 768 rows** before splitting; (iii) report **accuracy** as the main result even though the majority class is 65.1%; and (iv) present an ANN **without** the same-fold logistic and tree baselines that Smith et al. already considered necessary.

How this project answers that gap

1. Zeros are recoded as missing only for glucose, blood pressure, skin thickness, insulin, and BMI. Pregnancies stay at zero.
2. Imputation, scaling, `SelectKBest`, and optional oversampling live **inside** a pipeline fitted on each training fold (scikit-learn guidance on data leakage; Pedregosa et al., 2011).
3. The ANN is a small MLP compared with logistic regression and random forest on **identical** stratified folds.
4. The metric set includes sensitivity, specificity, precision, F1, ROC-AUC, PR-AUC, accuracy, and confusion matrices. Accuracy is reported, then immediately compared with the dummy majority baseline.

This is a **methods-and-evaluation** gap on a standard teaching dataset. It is not a claim of a new medical algorithm.

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3. Dataset and Data Understanding (8 marks)

3.1 Source and local file

Item	Detail
Name	Pima Indians Diabetes Database
Original owners	National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK)
Donor of the ML extract	Vincent Sigillito, Johns Hopkins Applied Physics Laboratory (received 9 May 1990 per the historical documentation)
Foundational paper	Smith et al. (1988)
Local files used	`diabetes_ann/data/pima_indians_diabetes.csv` and `data/pima_indians_diabetes.csv` (same extract). ARFF original: `data/diabetes.arff`
Download policy	The project **does not download** data. If the CSV is absent, the program stops and prints where to obtain it (UCI-historical / Kaggle search / convert local ARFF).

Warning. The current UCI repository entry **Diabetes, ID 34** (Kahn) is a **different** time-stamped insulin/glucose log. It is not this table. This project uses the 768-row, 8-predictor Pima extract.

3.2 Population and target

- **Who:** adult women, age 21 or older, of Pima heritage, community near Phoenix, Arizona.
- **Target (`Outcome`):** 1 = tested positive for diabetes under the study rule; 0 = tested negative.
- **Original clinical rule (Smith et al. / dataset notes):** WHO-style 2-hour OGTT glucose *geq 200* mg/dL, or the same threshold in routine care.
- **Public-file use:** rows are labelled 0/1. This course treats that label as the binary class to predict from the eight columns, not as a licence to diagnose new patients.

3.3 Features (eight predictors)

Feature	Meaning	Valid zero?
Pregnancies	Number of times pregnant	**Yes** (never pregnant)
Glucose	Plasma glucose at 2 hours in an OGTT (mg/dL)	**No** — 0 is not a plausible OGTT result
BloodPressure	Diastolic blood pressure (mm Hg)	**No** — 0 mm Hg is not a valid reading in living adults
SkinThickness	Triceps skinfold (mm)	**No** — 0 mm is not physiologically plausible
Insulin	2-hour serum insulin (μU/mL)	**No** — 0 is the missing-value code in this file
BMI	Body-mass index (kg/m²)	**No** — BMI 0 is not a valid measurement
DiabetesPedigreeFunction	Constructed family-history score (Smith et al.)	Zero is possible in the formula but none occur in this file
Age	Age in years	No zeros in this file; minimum is 21 by construction

3.4 Computed profile of the local CSV

These counts were computed from the local 768-row file. They match the historical documentation.

Check	Result
Rows × columns	768 × 9
Duplicate rows	0
True NaN in the file	0 in every column
Outcome = 0	500 (65.1%)
Outcome = 1	268 (34.9%)
Majority-class accuracy floor	~65.1% if the model always predicts “no diabetes”

Raw zero counts

Feature	Zeros	Action
Pregnancies	111	Keep
Glucose	5	Recode as missing
BloodPressure	35	Recode as missing
SkinThickness	227	Recode as missing
Insulin	374	Recode as missing
BMI	11	Recode as missing
DiabetesPedigreeFunction	0	Keep
Age	0	Keep

Missingness after that recode: Insulin 374/768 (48.7%), SkinThickness 227/768 (29.6%), BloodPressure 35 (4.6%), BMI 11 (1.4%), Glucose 5 (0.7%). Median imputation can fill these for modelling. It does **not** recover the true laboratory value.

3.5 Dataset limitations (must be said in the review)

1. **Small *n*.** 768 rows is small for a neural network with many weights.
2. **Narrow population.** Adult Pima women only.
3. **Label definition.** The original task was five-year onset under a 1980s WHO-style rule, not a 2026 clinic visit.
4. **Missingness disguised as zero.** Documentation that says “no missing values” is misleading.
5. **Heavy missingness** in insulin and skinfold after recoding.
6. **No external test community.**
7. **Not for diagnosis.**

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4. Data Preprocessing (5 marks)

Preprocessing is a **pipeline**, not a one-time edit of the CSV. Every statistical step that learns from values is fitted on the **training fold only**, then applied to that fold’s test rows (Pedregosa et al., 2011; scikit-learn “Common

pitfalls and recommended practices").

4.1 Step order (training fold)

1. **Recode implausible zeros to NaN** for Glucose, BloodPressure, SkinThickness, Insulin, BMI. Rule is fixed in advance; it does not use test-fold statistics. Pregnancies are not recoded.
2. **Median impute** (`SimpleImputer`). Median is learned from the training fold.
3. **Optional** `SelectKBest` (ANOVA *F*-test, default *k=5*). Scores are learned from the imputed training fold.
4. **Standard scale** (`StandardScaler`). Mean and variance from the training fold (after selection, if selection is on).
5. **Optional random oversampling** of the minority class (`RandomOverSampler`). Train fold only; never applied as a transform to the test fold.
6. **Classifier** (dummy, logistic regression, random forest, or MLP).

4.2 Outliers

IQR outlier counts were computed on **raw** values as a review, not as an automatic deletion rule. Raw zeros in blood pressure and insulin appear as outliers; that is expected and is handled by the zero-recode, not by dropping rows. Automatic deletion of IQR outliers on the full table is **not** used, because it would mix test information and shrink an already small sample. Residual outliers after recoding are left in place; tree models are relatively robust, and the MLP is small and regularised (*\alpha = 10^{-3}*, early stopping on a slice of the **training** fold).

Raw 1.5×IQR outlier counts (computed): Pregnancies 4, Glucose 5, BloodPressure 45, SkinThickness 1, Insulin 34, BMI 19, DiabetesPedigreeFunction 29, Age 9.

4.3 Scaling

Standardisation is required for the MLP and for logistic regression with a regularised linear solver. It is applied inside the pipeline so the ANN does not see raw mg/dL next to a 0–2 pedigree score. The random forest does not need scaling for splits, but it still sits in the same pipeline so the comparison stays fair.

4.4 Reproducibility

- Random seed **42** (Python, NumPy, and every `random_state`).
- Stratified 5-fold CV, shuffle on, same seed.
- Local CSV only; no silent download.
- Command: `cd diabetes_ann && python run_experiment.py --seed 42 --splits 5 --select-k 5`

Thread scheduling can cause tiny numeric drift in the MLP. If a number moves, compare `reports/metrics.csv` rather than a single printed accuracy.

4.5 Leakage prevention (explicit)

Practice	Allowed?
Imputer / scaler / selector fit on all 768 rows, then split	**No**
Oversample the whole table, then cross-validate	**No**
Choose *k* or hidden size by peeking at test-fold accuracy	**No**
Fit those steps inside `Pipeline` on each training fold	**Yes**
Recode zeros with a predeclared medical rule	**Yes**

5. Exploratory Data Analysis (7 marks)

EDA below uses the **local CSV**. Figures are in `diabetes_ann/reports/figures/`. Interpretation is descriptive. It is not a causal medical analysis.

5.1 Outcome distribution

500 negative vs 268 positive (65.1% / 34.9%). **Suggested visual:** `outcome_balance.png`.

Interpretation. Any model that always predicts "no diabetes" is about 65% accurate and 0% sensitive. Accuracy without a dummy baseline is not evidence of learning.

5.2 Feature distributions by outcome

Suggested visual: `feature_distributions.png`.

Computed means (raw file, zeros still present):

Feature	Mean, Outcome 0	Mean, Outcome 1
Glucose	110.0	141.3
BMI	30.3	35.1
Age	31.2	37.1
Pregnancies	3.30	4.87
Insulin	68.8	100.3
SkinThickness	19.7	22.2
BloodPressure	68.2	70.8
Pedigree	0.43	0.55

After recoding implausible zeros to missing, the Outcome-1 means for insulin and skinfold rise further (insulin 206.8 vs 130.3; skinfold 33.0 vs 27.2 among observed values). Glucose remains the largest raw gap.

Interpretation. Higher glucose, BMI, and age among positive rows is consistent with known diabetes risk factors and with Smith et al.'s variable choices. Insulin and skinfold comparisons are weaker in the raw file because of coded zeros.

5.3 Outlier review

Suggested visual: `outliers_boxplots.png` and `zero_counts.png` (red bars = treated as missing).

Boxplots on raw data show long right tails for insulin and pedigree, and a spike at zero for insulin, skinfold, and sometimes blood pressure. Those zeros are missing-value codes, not "unusually healthy" readings. They should not be interpreted as low-risk physiology.

5.4 Correlations

Suggested visual: `feature_correlations.png`.

Pearson correlation with `Outcome` **after** recoding implausible zeros to missing (computed):

Feature	*r* with Outcome
Glucose	0.495
BMI	0.314
Insulin	0.303
SkinThickness	0.259
Age	0.238
Pregnancies	0.222
DiabetesPedigreeFunction	0.174
BloodPressure	0.171

On the **raw** file, insulin (0.131) and skinfold (0.075) look much weaker because zeros dilute the association. That is an EDA reason for the recode, not a licence to fit the recode using the test fold.

Among predictors after recode, the strongest pairs are SkinThickness–BMI ($r = 0.648$), Glucose–Insulin ($r = 0.581$), and Age–Pregnancies ($r = 0.544$). Those pairs are redundant to a degree; that is why a within-fold selector is worth comparing with the full set, not because we assume a unique "best five features" for medicine.

5.5 What EDA does *not* show

- It does not show that an ANN will beat logistic regression.
- It does not show that the model works for men or for other ethnic groups.
- It does not replace a diagnostic test.

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6. Feature Engineering and Selection (7 marks)

6.1 Engineering choices (deliberately small)

No polynomial explosion and no target encoding. The table already contains a constructed feature (the pedigree function). Extra synthetic columns on 768 rows would overfit easily.

Engineering in this project means:

- medically justified missing-value recoding;
- median imputation;
- optional class rebalancing **on the training fold**;
- optional univariate selection **on the training fold**.

6.2 Feature-selection experiment

Question: does keeping the five ANOVA-*F* features chosen **inside each training fold** change CV metrics relative to using all eight features?

Protocol:

- Same stratified 5-fold splits for every model.
- Modes: `all_features` vs `kbest5`.
- Selector: `SelectKBest(score_func=f_classif, k=5)`, a pipeline step.

- Record the five names **per fold** (they are allowed to change).

Computed so far (local run, seed 42). Across models, fold-wise `SelectKBest` kept:

- **Every fold:** Glucose, BMI, Age
- **Often:** SkinThickness (4/5 folds), Pregnancies (3/5), Insulin (3/5)
- **Not selected in these folds:** BloodPressure, DiabetesPedigreeFunction

That pattern agrees in direction with the recoded correlations (glucose and BMI strongest) and with Howlader et al.'s qualitative note that glucose, BMI, and age are important — with the difference that **this project never selected those names on the full 768 rows first**.

Suggested visual: `feature_selection_frequency.png`.

6.3 Models (fair comparison)

Model	Role	Notes
Dummy most-frequent	Sanity floor	Always predicts the majority class
Logistic regression	Linear baseline	`class_weight='balanced'`, `liblinear`
Random forest	Tree baseline	100 trees, `max_depth=6`, balanced classes
MLP ANN	Feed-forward net	Hidden layers **(16, 8)**, ReLU, Adam, *\alpha=10^{-3}*, early stopping on 15% of the **training fold**

ANN diagram:

```
8 inputs → Dense(16, ReLU) → Dense(8, ReLU) → 1 logistic output
```

This is an **artificial neural network**, not a convolutional or recurrent net. scikit-learn's `MLPClassifier` is appropriate for a small tabular classroom problem (Pedregosa et al., 2011; scikit-learn MLP documentation). The library notes that `lbfgs` can be stronger on very small data; this implementation uses Adam plus early stopping. That is a stated design choice, not a claim of optimality.

Every model sees the **same folds** and the **same preprocessing**. The only intended difference is the classifier (and the on/off flags for selection and oversampling).

6.4 Evaluation metrics (why accuracy is not enough)

Reported on each test fold, then summarised as mean ± standard deviation:

Metric	Meaning here
Accuracy	Share of rows labelled correctly — misleading when 65.1% are negative
Precision	Of predicted positives, how many are true positives
Recall / sensitivity	Of true diabetes rows, how many are found
Specificity	Of true non-diabetes rows, how many stay negative
F1	Harmonic mean of precision and recall
ROC-AUC	Ranking quality across thresholds
PR-AUC	Ranking quality with emphasis on the positive class (better when positives are 35%)

Metric	Meaning here
Confusion matrix	Counts of TP, FP, TN, FN (also concatenated out-of-fold)

Smith et al. (1988) already reported **sensitivity, specificity, and ROC**, not accuracy alone. This project follows that clinical-style reporting even though the model itself is not a clinical device.

Dummy baseline (computed): accuracy 0.651 ± 0.002, recall 0, specificity 1, ROC-AUC 0.5. Any later ANN number must be read against that floor.

Full classifier tables are in `diabetes_ann/reports/model_comparison.md` for a later review. They are **not** a medical performance claim.

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7. Rubric coverage and missing information

This report addresses the **six visible Review I rows (40 marks)**. If the official sheet has further rows (implementation demo, viva, formatting, Review II model results), they were not in the screenshot text supplied and are **not** claimed as covered.

Please supply before submission

1. Guide, department, college logo / letterhead.
2. Official template (Word/PPT), citation style, page or slide limit.
3. Confirmation that Review I is 8 **October** 2026 (this package uses that date).
4. A photo or full sheet if the rubric continues below the six rows.
5. Any department ethics or “not a diagnostic tool” wording they require on the title slide.

The dataset file is **already present**. It is not on the missing list.

End of Review I report.